



Department of Education

**STEM Education
National Schools of Excellence
Teacher's Guide
Grade 12
Term 1**

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

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SECRETARY’S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle objective of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher’s Guide and Teacher’s Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Protein Sequencing	1. Amino Acids 2. Protein Structure and Function 3. Protein Purification 4. Protein Sequencing
HEALTH AND MEDICINE	12.1 Drug Discovery, Design and Development	1. Drug Targets 2. Drug Discovery 3. Computer Aided Drug Design 4. Drug Development and Production
ENERGY	12.1 Hydrogen – Future of Energy	1. Hydrogen as Fuel 2. Methods of Producing Hydrogen 3. Storing Hydrogen 4. Hydrogen Fuel Cell
BATTERY TECHNOLOGY	12.1 Solid-State Battery	1. Introduction to Solid-State Battery 2. Solid Electrolyte 3. Manufacturing process Solid-State Battery
ARTIFICIAL INTELLIGENCE	12.1 Artificial Molecular Machines	1. Introduction to Artificial molecular Machines 2. Catenanes and Rotaxanes 3. Other Protein based Motors under development
ROBOTICS	12.1 Robotics Meets Chemistry	1. Acidity Theories 2. Robotics 3. Programming a Robot

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – GRADE 12, TERM 1

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment				
Thema	Biotechnology	Module Content Standard		
Module 12.1 Protein Sequencing		12.1 To determine the amino acid sequence of a protein using various purification methods and sequencing methods.		
Topics	Performance Indicators	Module Benchmark	Assessment Tasks	
1. Amino Acids	12.1.1a: Explain what amino acids are 12.1.1b: Draw and describe the chemical structure of an alpha-amino acid. 12.1.1c: Name all 20 amino acids and identify them using the one-letter code and three-letter code. 12.1.1d: Classify the 20 amino acids into Polar, Non-polar, Basic and Acidic. 12.1.1e: Explain what peptide bond is 12.1.1f: Write the condensation reaction of two amino acids to form a dipeptide 12.1.1g: Differentiate between the N-terminus and the C-terminus	Students' ability to master the knowledge and skills of: - Protein Sequencing Process - Protein Purification Methods - Protein Sequencing Methods	Practical Assignment + Test + Quiz	
2. Protein Structures and Function	12.1.2a: Explain the four organizational levels of Protein Structure 12.1.2b: Differentiate between Polypeptides and Proteins 12.1.2c: Explain how chaperonins help in protein folding 12.1.2d: Explain what Denaturation of Protein is			

	12.1.2d: Justify how the structure of a protein determines the function of that protein		
3. Protein Purification	12.1.3a: List and Explains the different ways in which to purify proteins 12.1.3b: Explain why purification of proteins is an important step 12.1.3c: Describe the process of protein purification using Electrophoresis 12.1.3d: Purify a Protein using any one of those methods		
4. Protein Sequencing	12.1.4a: Define Protein Sequencing 12.1.4b: Explain the Process of Sanger's Method 12.1.4c: Explain the Process of Edman Degradation Method 12.1.4d: Utilize any one the protein sequencing methods to sequence a protein sample		

Thema	Health and Medicine	Module Content Standard	
Module 12.1 Drug Discovery, Design and Development		12.1: To present a plan on how one would take on drug discover, design and development process for a drug idea he/she has.	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Drug Targets	12.1.1a: Explain what drugs are and why we need them? 12.1.1b: List and discuss the different sources of drugs and lead compounds 12.1.1c: Explain the routes of drug administration 12.1.1d: Describe both the Pharmacokinetic phase and the Pharmacodynamic phase	Student's ability to master the knowledge and skills of: • Drug Actions • Drug Discovery Process • Drug Design using Computer Aided Design Programs • Drug Development Process and Patents.	Practical Assignment + Test + Quiz
2. Drug Discovery	12.1.2a: Explain the Process of Drug Discover 12.1.2b: Justify why Water Solubility is important 12.1.2c: Discuss what salt formation is 12.1.2d: Describe the process of incorporating water solubilizing groups in a drug structure		

3. Drug Development	12.1.3a: Explain the function of antibacterial agents 12.1.3b: Explain the function of antiviral agents 12.1.3c: Explain the function of anticancer agents 12.1.3d: Explain the function of cholinergic, anticholinergics and anticholinesterases 12.1.3e: Explain how drugs act on the Adrenergic Nervous System 12.1.3f: Explain the function of Opioid Analgesics 12.1.3g: Explain the function of antiulcer agents		
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Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standard	12.1	12.1
BIOTECHNOLOGY / 12.1 Protein Sequencing	12.1 To determine the amino acid sequence of a protein using various purification methods and sequencing methods.	✓	
HEALTH AND MEDICINE / 12.1 Drug Discovery, Design and Development	12.1: To present a plan on how one would take on drug discover, design and development process for a drug idea he/she has.		✓

THEMA: BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information)

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: PROTEIN SEQUENCING

4-5 Weeks

Protein sequencing is the technique of determining the amino acid sequence of a protein or peptide in its entirety or in part.¹ This module is designed to introduce students to the process or steps involved in Protein sequencing. The process involves:

1. Separating proteins into subunits
2. Producing Smaller Peptide Fragments
3. Sequencing the Protein Fragments
4. Repeating steps 2 and 3 using a different cleavage/hydrolysis method
5. Assembling the overlapping fragments for the full protein sequence

The entire practical learning experience will be covered within the following topics:

Topic 1: Amino Acids

Topic 2: Protein – Structure and Function

Topic 3: Protein Purification

Topic 4: Protein Sequencing

Linking Modules

- Biology / Thema: Biotechnology / Module 12.1: Bioinformatics
- Chemistry / Thema: Biotechnology / Module 11.1: Principles of Biotechnology / Topic 3: Proteins

Broad Content Standard

12.1: To determine the amino acid sequence of a protein using various purification methods and sequencing methods.

Content Benchmark

Students' ability to master the knowledge and skills of:

- Protein Sequencing Process
- Protein Purification Methods
- Protein Sequencing Methods

¹ (Protein Sequencing, 2021)

Topic 1: Amino Acids

Performance Indicators

Students are expected to:

- 12.1.1a:** Explain what amino acids are
- 12.1.1b:** Draw and describe the chemical structure of an alpha-amino acid.
- 12.1.1c:** Name all 20 amino acids and identify them using the one-letter code and three-letter code.
- 12.1.1d:** Classify the 20 amino acids into Polar, Non-polar, Basic and Acidic.
- 12.1.1e:** Explain what peptide bond is
- 12.1.1f:** Write the condensation reaction of two amino acids to form a dipeptide
- 12.1.1g:** Differentiate between the N-terminus and the C-terminus

Activity 1: Exercise

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Topic 2: Protein Structure and Function

Performance Indicators

Students are expected to:

- 12.1.4a:** Explain the four organizational levels of Protein Structure
- 12.1.4b:** Differentiate between Polypeptides and Proteins
- 12.1.4c:** Explain how chaperonins help in protein folding
- 12.1.4d:** Explain what Denaturation of Protein is
- 12.1.4d:** Justify how the structure of a protein determines the function of that protein

Activity 1: Self-Test Exercise

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Activity 2: Homework Questions

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Activity 3: Practical Experiment

Title: Laboratory Test of Proteins

In this activity, students are to perform some simple tests to identify the presence of proteins in the given sample.

(Refer to TRB)

Topic 3: Protein Purification

Performance Indicators

Students are expected to:

- 12.1.3a:** List and explains the different ways in which to purify proteins
- 12.1.3b:** Explain why purification of proteins is an important step
- 12.1.3c:** Describe the process of protein purification using Electrophoresis
- 12.1.3d:** Purify a Protein using any one of those methods

Student Task/Quiz/Test

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Topic 4: Protein Sequencing

Performance Indicators

Students are expected to:

- 12.1.4a:** Define Protein Sequencing
- 12.1.4b:** Explain the Process of Sanger's Method
- 12.1.4c:** Explain the Process of Edman Degradation Method
- 12.1.4d:** Utilize any one the protein sequencing methods to sequence a protein sample

Homework

Prepare the homework and hand to students to attempt

Resources: NIL

(Refer to TRB)

Assignment

Prepare the Assignment and hand to students to attempt

Resources: NIL

(Refer to TRB)

Practical Experiment 1

Aim: To perform some simple tests to identify the presence of proteins in the given sample.

Theory: Protein is a long chain polymer with a high molecular mass made up of -amino acids. Proteins are found in living organisms since they are components of cells. Carbon, hydrogen, nitrogen, oxygen, and phosphorus and sulphur are all found in proteins.

The following are the tests carried to find the presence of proteins in a given sample.

1. Biuret test
2. Xanthoproteic test
3. Million's test
4. Ninhydrin test

Resources:

1. Copper sulfate solution
2. Sodium hydroxide
3. Nitric acid
4. Mercuric sulfate
5. Sodium nitrite
6. Sulfuric acid
7. pyridine solution
8. Ninhydrin reagent
9. Distilled water
10. Test tubes
11. Test tube holder
12. Dropper
13. Water bath
14. Stirrer

Practical Experiment 2

Students will separate a mixture of proteins from skeletal muscle using SDSpolyacrylamide gel electrophoresis (PAGE). PAGE is a powerful analytical technique having numerous applications in modern biology. Evidence for evolutionary relatedness amongst organisms can be determined using this technique. Suggested organisms to compare include various fishes, mammals, poultry and/or sea foods, all of which are available from the grocery store. During the laboratory, students will develop an experimental design, prepare samples of muscle tissue from various organisms, observe the process of electrophoresis and analyze their results. Analysis consists of comparing protein bands according to their molecular weights on the gel

Information about Experiment

Teacher Prep: Significant, includes a trip to the grocery store Student Time: Either a full day in-school field trip or four 45 min. class periods; if the second option is chosen, either the teacher or students need to be available between classes and/or after school to turn off the gels and prepare them for staining at the end of the third day

Information with which students must be familiar:

- The structure and function of proteins, including what polypeptides are.
- Principles of electrophoresis.
- An understanding of evolution at the molecular level.
- Micro-scale units of measure

Required Time

- **Before class:** Ordering of proper equipment and supplies. 1-2 hours to make up all the solutions and put out supplies. A trip to the grocery store to obtain meat samples.
- **In class:** As written, this lab is designed to be done in one full day (4-6 hours) with one 45 min. class period for analysis and follow up. You can use the down time (e.g., when the gels are running and staining) for lecture/discussion on electrophoresis, proteins and/or molecular evolution, or you could do other labs or activities. You can also do the lab in four 45 minute class periods.
- Day 1: Background information and experimental design.
- Day 2: Extraction of muscle protein from animal tissues (the “master supply tubes” may be stored in the refrigerator overnight or in the freezer for longer time periods).
- Day 3: Gel electrophoresis.
- Day 4: Observation, analysis and follow up. If you have double periods (80-90 minutes) available for labs, you should consider combining the Extraction and Gel electrophoresis portions into one double period. If you choose one of these multi-day options, someone needs to be available between class periods and after school the day the gels are loaded to turn off the gels and do the staining procedure.

Materials

- Protein Gel Electrophoresis apparatus: (The number of electrophoresis setups required will vary depending on class size)
- mini vertical gel boxes w/ loading guides and buffer dams. Each box holds two gels.

- 1 thermometer
- power supplies and extra fuses
- staining trays with lids
- micropipettors (2-20 μ l) w/ gel loading tips
- 1 dry bath heater

Reagents and Gel Photography Supplies

- 1 bottle w/ ~ 90 mL 10x Tris-Glycine-SDS running buffer (per 4 gels)
- precast polyacrylamide gels (refrigerate upon arrival)
- 2 bottles blue Tris-SDS solubilization buffer, may be labeled “Laemmli Buffer” (per 4 gels)
- camera
- 2 bottles protein-fixative stain (per 4 gels)
- light box
- 1 vial of high molecular weight markers (freeze upon arrival) per 4 gels
- Polaroid film- enough film for one picture per gel, more film can be purchased from WARD’S Natural Science Establishment (1- 800-962-2660). Polaroid film type 667 can be purchased in packs of ten exposures (#25 W 1135, \$28.85).
- different meats
- distilled water (1 liter per gel)

General Lab Supplies

- 500 ml or 1 liter flask or beaker (one per gel box)
- labeling tape
- 100 ml graduated cylinder (at least one per class)
- 4 markers
- 500 ml graduated cylinder (at least one per class)
- razor blades or scalpels
- 15 ml blue-capped plastic tubes
- cutting boards and glass or plastic rods
- test tube racks
- 1 spatula
- clinical centrifuge or funnel with cheesecloth (one for whole class)
- safety glasses
- microcentrifuge tubes
- transfer pipettes (about 20)
- microcentrifuge racks
- latex gloves

Preparation Instructions

- The gels should be stored in a refrigerator, do not freeze the gels.
- Gloves should be used at all times when handling the gels.
- To prepare the gels the day of the lab, cut the gel packets open over the sink or a container, drain out excess buffer, and throw away the filter paper and plastic. Then use a razor blade to cut along the black line on the bottom front of the gel cassette, and peel off the strip of plastic covering the bottom front of the gel. This exposes

the gel to electric current. You can leave the gel cassettes at room temperature in a tray of buffer or distilled water until the students are ready to load them.

- The molecular weight markers (also referred to as the size standard) come in a 1.5 ml microcentrifuge tube, often labeled “Prot. std.” One tube has enough marker for 4 gels. You should store this tube in the freezer until the students begin preparing their gels for electrophoresis.
- When purchasing meats, keep in mind that you only need a very small amount of each sample. The deli section at your grocery store will usually give you small samples for minimal or no charge. Be sure to keep track of which meat is which. Note: you need two sets of 15 ml conical tubes, one set for steps 4-7 of the sample preparation, and a new set for steps 8-10!
- All chemicals should be prepared according to the manufacturer’s directions. The Tris-Glycine-SDS running buffer is supplied as a 10X concentrate. The buffer will be sudsy because it contains SDS. The students can add 40 ml of this 10X buffer and 360 ml of deionized water in a 500 ml graduate cylinder to produce 400 ml of 1X buffer, which is enough for one gel box. The bottle contains about 90 ml of 10X running buffer, enough for four gels.
- The blue Tris-SDS solubilization buffer (often labeled Laemmli Buffer) is a 4X stock solution. The students dilute it to 1X when they add 3 ml of deionized water to their samples; do not dilute this before hand.
- The stain is not reusable. You should pour it down the sink with water.

Safety Precautions

Good laboratory practice should be followed while carrying out all aspects of this lab. Students should be required to wear gloves and safety glasses while handling the meats, the polyacrylamide gels, the protein stain and the other chemicals. Be sure that students wash their hands and bench tops after working with the raw meats, especially if you purchase chicken or seafood

Helpful Tips

- The ideal group size is four students per gel, but 2-6 students will also work.
- All of the materials in the CIBT kits are available commercially. You can order them all from Bio Rad or Ward’s. The gel boxes and gels can be ordered from Bio Rad. If you order materials from another source, be sure to make necessary modifications due to differences in equipment/product.
- This lab requires large amounts of distilled/deionized H₂O. You can have each student bring in a gallon of distilled water, and you can use the excess other times of the year.
- The questions the students ask will depend on the meats available. In addition to grocery store meats, other samples can be used. For instance, one teacher went to a French restaurant and obtained frog legs. Possible questions/ experimental designs include:
 - They call pork “the other white meat”? Is its protein composition more similar to chicken or to beef?
 - What is imitation crab made of? (For this you might want to have students compare imitation crab to chicken, several species of fish and crab legs. Crab legs are not available year round; you can substitute

shrimp and tell students that shrimp, like crab is a crustacean. If imitation crab is made from something similar to crab it should have a protein composition similar to shrimp.)

- Broad level phylogenetic comparisons
 - Similarities and differences between fresh water and salt water fish
 - Land animals vs. sea animals
 - Vertebrates vs. invertebrates
 - Provide the students with a mystery meat and have them classify it. Avoid highly-processed or cooked meats; they may contain non-muscle proteins.
- There are 15 lanes available for loading. However, if you use the loading guides, only lanes 2-14 are easy to load. It works well to have each group choose 5-7 samples. This provides the opportunity to load each sample twice in case anything goes wrong with the loading. Make sure at least one lane contains molecular weight markers but pick a lane off to the side to help to orient the gel after it gets flipped during removal and staining/destaining.
 - Please refer to the addendum “Gel Box Assembly” when setting up the gel boxes and inserting the gels. It’s especially important to remember to press down on the top of the electrode assembly while closing the two cam levers on the clamping frame. Otherwise buffer may leak from the upper chamber, interrupting electrophoresis.
 - Students can indicate the location of each well on the long gel plate with a marker. If you are still having trouble seeing the wells, look at the gel box against a light background. Once one well is loaded, it’s much easier to find the others. Students can also write their names on the long plate to identify the gels. If you choose to use the loading guides, remember they are only guides.
 - Remove the comb from the top of the gel using two hands just before loading. Use a transfer pipette to rinse out the wells.
 - Samples should be pipetted carefully but quickly into the wells at the top of the gel. Since there are two gels to a box, one group of students may have to wait to load their gel until the first group is finished. If too much time is spent between loading the first sample and turning on the current, diffusion of proteins will occur and the gel could be a mess. Don’t move the gel boxes after samples have been loaded.
 - If you are running an odd number of gels, insert one of the buffer dams instead of a gel into the other side of the clamping frame, so that the inner buffer chamber will be sealed.
 - Some meats, such as clam or squid, are hard to mash. Mincing the samples into very tiny pieces is the key to success.
 - A 15% polyacrylamide gel used in this lab should run for 40-50 min to show the best distribution of intact proteins.
 - Gels will destain much more quickly and effectively if 30% methanol is used as the destain. Shaking or rocking the gels will speed up the destaining process.
 - Caution should be used when working with the gels during staining/destaining. The gels are very fragile and rip easily.
 - The size standard is measured in daltons. A Dalton (Da) is the unit of measure of molecular weight. One dalton is equivalent to the weight of one hydrogen atom. One amino acid weighs, on average, 120 daltons. So, you can approximate the number of amino acids in a protein based on its molecular

weight. You can introduce this concept to more advanced students and use it in pre-lab question #2 and the optional post-lab question #7.

- Advanced students would benefit from doing the CIBT lab Biological Shapes prior to doing this lab.
- If you have covered the proteins in motor muscle, you can discuss that topic in relation to this lab. For example, you should see a band at 46 kDa in all of your muscle samples. This band represents actin. All animals, both invertebrate and vertebrate, have actin as a primary component in their muscle. Myosin, the second major muscle protein, is very large and does not separate well on the gels we are using. It is found in the prominent band at the top of every lane.

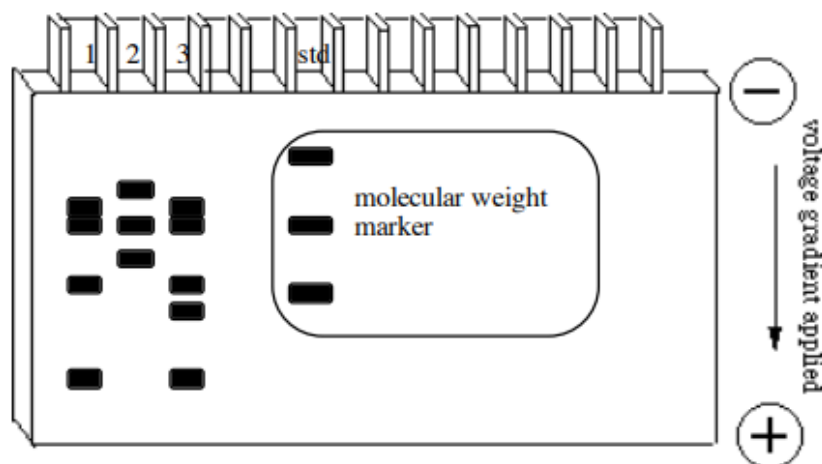
Questions and Answers

1. Why do SDS-coated polypeptides separate when placed in an electrical field?

Because they are coated with SDS, polypeptide samples prepared for SDS gels have a net negative charge. When they are placed in the electrical field, they will migrate toward the positive electrode. The gel is a restrictive material through which small molecules will pass more readily than large molecules. Polypeptides of higher molecular weight will therefore be found closer to the wells than smaller molecules which will move further down the gel during electrophoresis.

2.

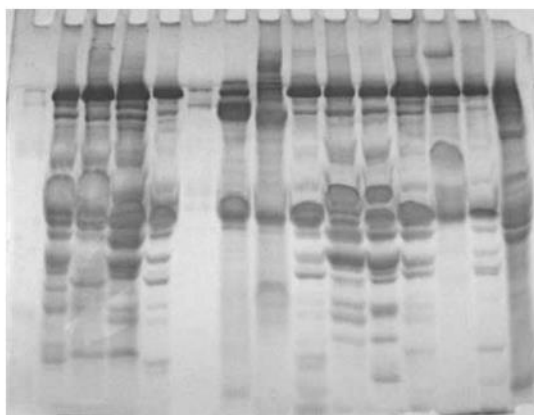
- a. On the diagram below, sketch the predicted results if lane #1 was loaded with a sample composed of 4 different types of polypeptides (having 100 amino acids, 175 amino acids, 250 amino acids, 300 amino acids), lane #2 was loaded with a sample composed of 3 types of polypeptides (200 amino acids, 250 amino acids, and 350 amino acids) and lane #3 was loaded with a sample of 5 different types (100 amino acids, 150 amino acids, 175 amino acids, 250 amino acids, 300 amino acids). Note: The average amino acid has a molecular weight of 120 daltons. Therefore, more advanced students could be asked to compare the information above to a molecular weight standard in lane 4, the picture below includes the molecular weight standard.
- b. If each lane represents the proteins from a different species of animal, which two would you say are the most closely related?
Why? Organisms 1 and 3, because their protein composition is the most similar. They share a large number of bands.



3. Restate the question your group was asking.
Answers here will vary depending on the experimental design.
4. Describe or sketch the appearance of the gel initially.
In each well there is some blue material that is visible on top of the gel. In some wells this blue material is very faint.
5. Describe or sketch the appearance of the gel after 5 minutes. Explain the difference between the “just loaded” gel and the gel after 5 minutes.
After five minutes the blue material has moved down into the gel. It is now below the black rubber gasket. The protein dye has a negative charge and has started moving towards the anode.
6. Observe the bands on the gel and make a sketch of your gel. Be sure to indicate the positive and negative sides of the gel and label each lane indicating which sample was loaded into it.
Answers here will vary depending on the protein samples used.
7.
 - a. Based on your analysis of the gel, which organisms are closely related?
Answers here will vary depending on the protein samples used.
 - b. Which organisms are distantly related? How do you know?
Answers here will vary depending on the protein samples used. Closely related samples will have many bands located at the same location at a similar concentration.
8. Discuss the results of your experiment in light of the questions you have asked?
Answers here will vary depending on the experimental design.
9. It is possible to determine the size of individual proteins by comparing their distance migrated on a gel to the distances moved by proteins of known size. The distance migrated is inversely related to the log of the size.

- a. From your gel photo (or from the gel itself) measure the distance from the well to each of the bands in the lane corresponding to molecular weight markers.
- b. Obtain a piece of 2 cycle semi-log paper. For each band, plot the distance migrated from the well on the linear axis and the size (molecular weight) on the log scale. Connect the points with a smooth curve. This is a standard curve and can be used to determine the size of any of the proteins separated from your muscle tissues. Conditions vary from gel to gel, this is why it is very important to include molecular weight markers on every gel that you run. You cannot use a set of molecular weight markers from another gel in your class to determine the size of proteins on your gel. Determine the size of several of the most abundant proteins in each of your samples.
- c. An amino acid is approximately 120 daltons. Use this information to calculate the approximate number of amino acids in the proteins you sized in part b.

A 15% gel run for 50 minutes, stained 4 minutes and destained in 30% methanol. Note the pronounced bands for myosin (203,000 Da) and actin (46,000 Da) in every lane.



(Refer to TRB)

SAMPLE ASSESSMENT

1. Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

2. Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks	
12.1: Protein Sequencing	12.1 To determine the amino acid sequence of a protein	1. Amino Acids	Written Quiz	Project + Written Test
		2. Protein Structure and Function		
		3. Protein Purification	Written Quiz	
		4. Protein Sequencing		

3. Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1. Amino Acids	Written Quiz	Explain what amino acids are	2.5
		Draw and describe the chemical structure of an alpha-amino acid.	2.5
		Name all 20 amino acids and identify them using the one-letter code and three-letter code.	5
		Classify the 20 amino acids into Polar, Non-polar, Basic and Acidic.	5
		Explain what peptide bond is	2.5
		Write the condensation reaction of two amino acids to form a dipeptide	2.5
Subtotal			20
Module 12.1 (All four Topics)	Project	Purify Protein Sequencing	20
		Determine the amino acid sequence of a protein	20
		Check if the protein sequence exist on a database of protein sequences online.	10
Subtotal			50
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	15
		Correctly answers short answer questions	5
		Subtotal	30
Module TOTAL			100

THEMA: HEALTH AND MEDICINE

Introduction

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

The study of the human and body, how this function, and how they interact, not only with each other but also with their environment, has been of utmost importance in ensuring human well-being. Research on potential treatments and preventive medicine has expended greatly with the development of modern medicine, and a network of disciplines, including such fields as genetics, psychology, and nutrition, aims to facilitate the betterment of our health.

MODULE 12.1: DRUG DEVELOPMENT PROCESS

4-5 Weeks

Papua New Guinea (PNG) has the potential to solve some of the world's health-related problems regarding therapy and treatments drugs. Why? Because PNG has majorly unexplored rainforests and large biodiversity. Think about it! PNG has a rich tradition of using herbal medicine, and about 50% of our rural population relies on traditional herbal medicine for therapy and treatment. This should imply to us that PNG is a potential source for drug discovery².

We (Students) need to be encouraged to exploit the scientific opportunities that our country has. If we choose to investigate that comparative advantage, we may likely stumble upon what may be new medical technologies or drugs that may not only solve PNG's health-related problems, but the worlds.

Medicinal Chemistry is an interdisciplinary study that combines aspects of chemistry, physical chemistry, pharmacology, microbiology, biochemistry, and computational chemistry. Medicinal Chemistry to the micromolecular level is a very interesting yet complex topic. However, for this Module, the focus will be on the process of discovery and development of drugs at a bird-eye view. By the end of this Module, the students should be sufficiently equipped with the concept of drug discovery and development should they choose to conduct further research on drug development as well as pursue a career in the field of pharmaceutical biotechnology.

Linking Modules

Prerequisites and links:

- Biology / Thema: Health and Medicine / Module 11.1: Health and Medicine Sciences
- Chemistry / Thema: Health and Medicine / Module 11.1: Biochemistry in Medicine

Broad Content Standard

12.1: To present a plan on how one would take on drug discovery and development process for a drug idea he/she has.

Content Benchmark

Student's ability to master the knowledge and skills of:

- Drug Actions
- Drug Discovery Process
- Drug Development Process and Patents.

² [2327-5162.S1.011-006.pdf \(hilarispublisher.com\)](#)

Topic 1: Drug Targets

Performance Indicators

Students are expected to:

- 12.1.1a:** Explain what drugs are and why we need them?
- 12.1.1b:** List and discuss the different sources of drugs and lead compounds
- 12.1.1c:** Explain the routes of drug administration
- 12.1.1d:** Describe both the Pharmacokinetic phase and the Pharmacodynamic phase

Activity 1: Group Presentation

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover background on the drug, brief discovery history of the drug, what the drug targets, and its drug action (pharmacokinetics and pharmacodynamics)
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Resources: NIL

(Refer to TRB)

Activity 2: Practical (Optional)

Title: Evaluating Drug Action in Humans

This experiment aims to have students study drug action in humans. Below are the overall objectives of this experiment:

- To understand the importance of informed consent in the conduct of experiments on humans.
- To investigate the variability in baseline parameters in a group of normal healthy volunteers.
- To investigate the differences between β -adrenoceptor antagonists on cardiovascular and
- respiratory parameters using a semi-automated blood pressure monitor, peak flow meter and
- skin temperature thermistor.
- To assess the validity of indirect measures to determine drug effects in humans.
- To gain an understanding of the use of abstracts as a form of scientific communication.

Resources: Placebo, OR Atenolol (100 mg) OR Pindolol (10 mg)

(Refer to TRB)

Activity 3: Exercise

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Topic 2: Drug Discovery

Performance Indicators

Students are expected to:

- 12.1.2a:** Explain the Process of Drug Discover
- 12.1.2b:** Justify why Water Solubility is important
- 12.1.2c:** Discuss what salt formation is
- 12.1.2d:** Describe the process of incorporating water solubilizing groups in a drug structure

Activity 1: Group Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover the drug discovery phases.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Resources: NIL

(Refer to TRB)

Activity 2: Practical

Title: Synthesizing Oil in Wintergreen from Aspirin

In this experiment, students will synthesize Oil from Aspirin and study the drug development process.

Resources: Methanol, sulfuric acid, salicylic acid, and ethanol were purchased from Sigma-Aldrich; Filter paper, large (90 cm diameter); Petri dishes, agar, 6-well plates, Aspirin (regular strength 325 mg tablets, noncoated, nonbuffered), cotton swabs, bleach, Gloves, Mortar and pestles, Bacteria stab, Sterile water, fume hood, water bath, culture incubator.

(Refer to TRB)

Activity 3: Exercise

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

Topic 3: Drug Development and Production

Performance Indicators

Students are expected to:

- 12.1.3a:** Explain the function of antibacterial agents
- 12.1.3b:** Explain the function of antiviral agents
- 12.1.3c:** Explain the function of anticancer agents
- 12.1.3d:** Explain the function of cholinergic, anticholinergics and anticholinesterases
- 12.1.3e:** Explain how drugs act on the Adrenergic Nervous System
- 12.1.3f:** Explain the function of Opioid Analgesics
- 12.1.3g:** Explain the function of antiulcer agents

Activity 1: Group Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover the development phase of the drug and its performance in the pharmaceutical industry/market.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 1: Homework/Assignment Exercise

Prepare the exercises and hand to students to attempt

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

1. Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

2. Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Medical Chemistry (Drug Discovery, Design and Development)	12.1: To present a plan on how one would take on drug discovery and development process for a drug idea he/she has.	1. Drug Targets	Written Test
		2. Drug Discovery	
		3. Drug Development and Production	

3. Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
Module 12.1 (All Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	15
		Correctly answers short answer questions	5
		Subtotal	30
Module TOTAL			30

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules	Total Marks	% Weight of Modules
	6		
Theory Knowledge & skills – Test	Test (50)	/50	20
Application of knowledge – Assignments and projects	Assign.1 (30)	/30	30
End of Term Exam	(25 marks)	/25	50
Total marks for term one			100 marks

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Name s					Assessment Task				Total Score (/300)	Percentage Weight (%)				Achie vemen t Level [VHA, HA, SA,LA]
		Tests		Assignments				Project	Exam (/50)		T (20%)	A + P (30%)	Exam (50%)	Ttl CA	
1	Bill Joe	45	47	20	21	22	23	55	45	278	18.4	25.88	45	89.2 8	VHA
2	Cara Den	20	25	15	16	17	18	25	30	166	9	18.22	30	57.2 2	SA
3	Lea May	9	15	5	6	7	8	30	25	105	4.8	9.10	25	38.9	LA

Assessment Reporting

Thema / Biotechnology / Module 12.1: Protein Sequencing Broad Content Standard for Assignments / project 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Determine Protein Sequence (15 marks)	Prepare Lab Report (5 marks)	Answers all Relative Questions to protein sequencing (5 marks)	25 marks
Very High Achievement	Accurately determine protein sequence (13-15)	Accurately writes a lab report with correct sections (5)	Accurately answers all relative questions to protein sequencing (5)	(21-25)
High Achievement	Partially determine protein sequence (10-12)	Partially writes a lab report with correct sections (4)	Partially answers all relative questions to protein sequencing (4)	(16-20)
Satisfactory Achievement	Fairly determine protein sequence (7-9)	Fairly writes a lab report with correct sections (3)	Impartially answers all relative questions to protein sequencing (3)	(11-15)
Low Achievement	Poorly determine protein sequence (4-6)	Poorly writes a lab report with correct sections (2)	Poorly answers all relative questions to protein sequencing (2)	(6-10)
Below minimum Achievement	Needs improvement in determining protein sequence (0-3)	Needs improvement in writing a lab report with correct sections (1)	Needs improvement in answering all relative questions to protein sequencing (1)	(0-5)

Thema Biotechnology / Module 12.1: Protein Sequencing				
Broad Content Standards For Test: 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Correctly answers all long questions (25 marks)	Correctly answers all multiple choice (10 marks)	Correctly use concepts to design and propose solutions(15 marks)	50 marks
Very High Achievement	Accurately answers all long questions (21-25)	Accurately answers all multiple choice (9-10)	Accurately use concepts to design and propose solutions (13-15)	(41-50)
High Achievement	Partially answers all long questions (16-20)	Partially answers all multiple choice (7-8)	Partially use concepts to design and propose solutions (10-12)	(31-40)
Satisfactory Achievement	Fairly answers all long questions (11-15)	Fairly answers all multiple choice (5-6)	Fairly use concepts to design and propose solutions (7-9)	(21-30)
Low Achievement	Poorly answers all long questions (6-10)	Poorly answers all multiple choice (3-4)	Poorly use concepts to design and propose solutions (4-6)	(11-20)
Below minimum Achievement	Needs improvement in answering all long questions (0-5)	Needs answers all multiple choice (0-2)	Needs improvement in using concepts to design and propose solutions (0-3)	(1-10)

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